
World Properties without World Models: A Static-Embedding Baseline for Claims about the Internal States of Language Models

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Abstract

A rapidly growing literature shows that nameable variables can be linearly decoded from the hidden-state activations of large language models (LLMs). These range from properties of the world, such as the locations of cities and the lifetimes of historical figures, to emotions and pain. Such findings are often taken as evidence that language models go beyond surface text statistics and form internal models of the world, and even of themselves. Here, we show that static word embeddings of the same or matched stimuli, which are fixed, context-insensitive lexical representations learned from corpus statistics, support much of the same decoding. Across four published cases, static vectors predict place coordinates and year of death ($R^2 = 0.42\text{--}0.59$), separate pain from matched control sentences (held-out AUC 0.85–0.88), and classify twelve emotions in stories that were written specifically to avoid naming them (AUC 0.84–0.88). For the emotions, the vector for the emotion word alone captures a substantial share of the signal. Because static embeddings are a crude way of representing word associations, these results are a lower bound on what associations alone can support. The LLMs retain clear advantages on several representational tests, sometimes substantial, and causal and behavioral findings remain outside the scope of the baseline. On the original authors’ entities, where we reproduce their Llama-2 results, the transformer’s advantage lies mostly in placing historical figures in the right century and places in the right country, and within those groups every representation orders items poorly. These results do not show that language models lack representations of geography, time, pain or emotion. They show that decodability alone does not distinguish such representations from information already present in context-insensitive word statistics, and we suggest that any decodability result be checked against a static-embedding baseline on the same stimuli before it is taken as evidence about the model.

1 Introduction

Over the past several years, there have been numerous claims that large language models build internal representations of properties that go beyond the statistics of word relations. Most of these claims rest on decoding, in which a simple (typically linear) model is trained to predict a high-level property of input items, such as where a city is, when a person lived or what a character feels, from the language model’s internal activations. If the decoder succeeds on items held out from its training, the language model is taken to represent the property. Some of the most prominent examples decode properties of the external world, or internal, experience-like states. For example, decoders trained on the hidden states of Llama-2 predict the rough geographic coordinates of places and dates associated with people, artworks and news headlines, which Gurnee & Tegmark (2024) describe as “the basic ingredients of a world model.” Sofroniew et al. (2026) show that the activations of Claude Sonnet 4.5 predict which emotion a character in a story is feeling, and Tagliabue et al. (2026) find that the activations of 25 open-weight models predict whether a sentence describes pain. Both also report that these representations shape behavior, from the model’s rates

of blackmail and sycophancy to steered models choosing actions that delete a user’s photos or their own weights.

The conclusions drawn from this method are often strong. Gurnee & Tegmark (2024) ask whether LLMs “just learn an enormous collection of superficial statistics or a set of more coherent and grounded representations that reflect the real world,” and “find evidence for the latter.” Sofroniew et al. (2026) “interpret these representations as encoding the broad concept of a particular emotion, generalizing across the many contexts and behaviors it might be linked to.” Tagliabue et al. (2026) set out to test “whether LLMs represent pain distinctly from fear, sadness, and generic negative valence,” and title their paper “The Pain Axis: LLMs Represent Self-Directed Harm and Act on It.” Such results are now cited as relevant to questions of model sentience and welfare (Keeling et al., 2024).

Behind these conclusions is a shared intuition that a large network, trained on vast amounts of text, comes to model what the text is about, abstracting across many layers of processing from the words themselves to the places, times and states they describe. We use *world model* in this sense: structure in a model’s activations that is organized by the properties of what is described, such as a map or a timeline, rather than by the words used to describe it. There is, however, a potentially simpler explanation for these results, one that goes back to the observation that “you shall know a word by the company it keeps” (Firth, 1957, p. 11): words that relate to the same concepts tend to be surrounded by the same other words. Places in the same region are likely to share terminology regarding language, culture and weather, while historical figures tend to be described in the vocabulary of their eras, with certain words concentrated in certain epochs. The same is true for inner states: a story about grief shares vocabulary with other stories about grief even when the word *grief* never appears. An LLM, trained to exploit these word relations in order to generate the next token from the context, will therefore likely give similar internal representations to neighboring places, contemporaries and similar feelings, placing them near one another in its activation space. A decoder trained on those activations could therefore succeed without showing that the model has constructed a map, a timeline or an affective state beyond these word associations.

The question we ask is therefore whether word associations alone can produce decoding results like those reported for language models. The answer matters because successful decoding from fixed lexical associations would not distinguish a context-dependent world model from information already available in corpus-derived lexical structure. If such associations account for much of what is decoded, decoding alone cannot support the stronger world-model interpretation.

Static word embeddings offer a direct test of this explanation, because they can do none of the things that intuition depends on. Each word’s vector is a compressed record of which other words it occurs with, computed once from a corpus. Once the vectors are trained, no network processes the stimulus and there is no layered, context-dependent computation: a name, sentence or story corresponds to nothing more than the average of its words’ vectors, with their order discarded. In the phrasing of Gurnee & Tegmark (2024), static embeddings are “an enormous collection of superficial statistics” and nothing more. Whatever they predict about a property is therefore carried by word relations alone, with no contextual computation to build anything more. They are also a deliberately crude way of representing word associations, so what they achieve is a lower bound on what associations can support. Language models represent the same associations far better: they learn from vastly more text and longer contexts, and, above all, a word’s relations to other words are not fixed across its uses but shaped by the context in which it currently appears. In a static embedding, *sharp* relates to other words in the same way in *a sharp pain* and *a sharp suit*; in a transformer it need not. A gap between static embeddings and language models is therefore expected even if contextualized lexical and distributional structure accounts for the decoded signal. What the comparison can show is how far a crude implementation of word association already goes, and whether the models’ advantage looks like better word association, such as finer sorting and sensitivity to how words combine in a sentence, or like structure organized by the properties themselves, such as a map that extends to regions the decoder has not seen.

Here, we take four published cases and rerun each analysis with static embeddings of the same stimuli in place of the language model’s activations, keeping the same decoders, splits, controls and metrics, and compare the results with those reported for the language models. For three of the cases we use the original stimuli;

for emotions, whose stimuli are not released, we use stories generated under the same protocol, so that case is a conceptual replication rather than a same-stimulus comparison. We use GloVe (Pennington et al., 2014), Word2Vec (Mikolov et al., 2013a) and fastText (Bojanowski et al., 2017).

We find that static embeddings reproduce much of what has been taken as evidence for these representations in all four cases. They predict place coordinates and year of death on the datasets released by Gurnee & Tegmark (2024); they separate pain from matched controls on held-out sentences, well above the keyword baselines taken to rule out a lexical explanation, and reproduce how the pain direction’s overlap with fear and negative emotion depends on how the directions are constructed; and they classify twelve emotions in stories written never to name them, with the vector for the emotion word alone capturing a substantial share of the signal. The language models perform better on several of these tasks, sometimes substantially, and the advantage is consistent with a better representation of the same word associations, adjusted to context and learned by far larger models; it does not require positing a qualitatively different representation of the property itself. Where we can examine the advantage directly, including on the original authors’ entities, it lies mostly in coarse sorting, placing historical figures in the right century and places in the right country, rather than in the finer-grained ordering within them that a map or timeline would provide, where all representations are weak.

2 Related Work

Probing began as a diagnostic of what information a layer makes available (Alain & Bengio, 2016; Conneau et al., 2018; Belinkov, 2022) before it came to be used as evidence of what a model represents. Beyond the cases we examine, decoding has been used to argue that language models represent syntax (Hewitt & Manning, 2019), the state of entities in a discourse (Li et al., 2021), the state of an Othello or chess board (in models trained on game moves) (Li et al., 2023; Nanda et al., 2023; Karvonen, 2024), color (Abdou et al., 2021; Patel & Pavlick, 2022), numeric attributes such as birth year and population (Heinzerling & Inui, 2024), truth and falsehood (Burns et al., 2023; Azaria & Mitchell, 2023; Marks & Tegmark, 2023), what a character in a story believes (Zhu et al., 2024), attributes of the user (Chen et al., 2024), character traits (Chen et al., 2025), deception (Goldowsky-Dill et al., 2025), and sentiment and emotion (Tigges et al., 2023; Zou et al., 2023; Sun et al., 2026).

Static embeddings and controls for probes. It is already well established that word co-occurrences carry information about high-level structure. Latent semantic analysis recovers the relative locations of cities (Louwerse & Zwaan, 2009), even in “Middle Earth” (Louwerse & Benesh, 2012); regression from Word2Vec vectors predicts the coordinates, population and GDP of countries and cities (Gupta et al., 2015; Konkol et al., 2017); projections onto antonym directions recover human ratings of size, danger and temperature (Grand et al., 2022), and of valence and arousal (Hollis et al., 2017); and the difference-of-vectors directions approach which is now standard for LLMs was first introduced on static embeddings (Mikolov et al., 2013b; Bolukbasi et al., 2016; Kozłowski et al., 2019). The closest precedent to the current work, Liétard et al. (2021), found BERT, RoBERTa and GPT-2 no better than Word2Vec at predicting city coordinates. Gurnee & Tegmark (2024) cite this work but critique it as limited to a few hundred cities and small models, and ultimately do not test static embeddings on their own data. Probes themselves have long been known to succeed by exploiting superficial regularities (Hewitt & Liang, 2019), and what a probe can decode need not be what the model uses (Ravichander et al., 2021; Elazar et al., 2021; Belinkov, 2022); truth probes fail to generalize in basic ways (Levinstein & Hermann, 2024; Farquhar et al., 2023), and emotion directions have been proposed to act through the lexicon (Sun et al., 2026). Our contribution is not the observation that static embeddings carry such structure, but its use as a control on specific published claims. First, we rerun the published analyses with static embeddings in place of model activations, in three direct same-stimulus reanalyses and one protocol-matched conceptual replication, which measures how much of the evidence offered for a representation is already present in word relations. Second, where the language models do better, we separate their advantage into coarse sorting between groups, such as continents and centuries, and finer ordering within them, and compare each with static embeddings. Third, we extend the comparison from properties of the world to claims about internal states, where no such control has been applied. Fourth, we show that the controls these studies used to rule out a lexical explanation, such as keyword baselines and

stimuli that never name the target state, are passed by static embeddings, and so do not rule it out. More generally, we argue that static embeddings should serve as a standard baseline for any claim that a language model represents a property of what its input describes.

3 Methods

3.1 Static Embeddings

We use three pretrained static word embeddings, each with 300 dimensions:

- **GloVe 6B** (Pennington et al., 2014): trained on 6 billion tokens from Wikipedia 2014 and Gigaword 5, with a 400,000-word vocabulary. GloVe fits a log-bilinear model to the log of word co-occurrence counts.
- **Word2Vec Google News** (Mikolov et al., 2013a): trained on about 100 billion tokens of Google News, with a 3-million-item vocabulary that includes phrases such as `New_York`.
- **fastText wiki-news** (Bojanowski et al., 2017): the released 1-million-word vector file (`wiki-news-300d-1M.vec`), trained with subword information; because we use the exported vectors rather than the binary model, words outside its vocabulary get no vector. Used for the released entities and the pain and emotion cases.

Text is split into words (runs of letters and digits, keeping internal apostrophes and hyphens). GloVe is looked up in lower case; Word2Vec and fastText are looked up as written, falling back to lower case. A multi-word name, sentence or paragraph is represented by the unweighted mean of the raw vectors of its in-vocabulary words; items with no in-vocabulary word are excluded from the entity analyses (Table 7 in the appendix gives coverage). For entity names, Word2Vec’s phrase entries (e.g. `New_York`) are used where they exist; replacing them with averaged words changes place and death-year R^2 by at most 0.01. The three differ in how they are trained (GloVe fits co-occurrence counts, Word2Vec uses a predictive objective, fastText adds subword information), but what matters here is what they share: each is a fixed, context-insensitive lexical representation, with no computation over the stimulus beyond averaging.

3.2 Probes

Continuous targets are predicted with ridge regression (Hoerl & Kennard, 1970). The regularization strength is chosen within the training set by generalized (efficient leave-one-out) cross-validation, as implemented in scikit-learn’s `RidgeCV`, over twenty values from 10^{-2} to 10^3 (29 values from 10^{-2} to 10^5 for the released entities and transformer activations). Probes include a fitted intercept, since several targets (e.g. year of death) have means far from zero.

In each case we use the same linear decoder as the original study: ridge regression for place and time (Gurnee & Tegmark, 2024), and projection onto a difference-of-means direction for pain and emotion (Tagliabue et al., 2026; Sofroniew et al., 2026), described in Sections 3.6 and 3.7. Gurnee & Tegmark (2024) report that nonlinear decoders add little over linear ones. In every cross-validated analysis, directions, denoising components and standardization are fit on training folds only. The comparison is between representations under the same decoder and on the same entities; it cannot distinguish the contributions of corpus scale, training objective, architecture, contextualization and dimensionality to any difference between them.

3.3 Datasets

Released entity datasets. For the main spatial and temporal comparison we use two of the six entity datasets released by Gurnee & Tegmark (2024): 39,585 places worldwide with latitude and longitude, and 37,539 historical figures with birth and death years, each with the authors’ designated train–test split. We do not analyze their places in the United States and New York City, or the release dates of artworks and publication dates of news headlines. Our claims concern their probe results, not their prompting, cross-entity or neuron-level findings.

World cities (N=272). For targets the release does not include (temperature, population, GDP per capita and elevation) and for the residualization and ablation analyses, we use 272 cities spanning six continents (36 in Africa, 74 in Asia, 59 in Europe, 70 in North America, 13 in Oceania, 20 in South America) and latitudes from -43.5° (Christchurch) to $+64.8^\circ$ (Fairbanks), all present in both GloVe and Word2Vec. The set was assembled informally in earlier stages of this project to span regions and climates, without a fixed sampling rule, and we cannot document how individual cities were chosen or whether some were added after earlier probe results had been seen. It is restricted to names in both vocabularies and favors prominent, well-documented cities. It is therefore a convenience sample: the numerical estimates on it apply to this set rather than to a representative sample of world cities, and our conclusions about the published claims rest mainly on the released entities, which we analyze with the same methods. Coordinates come from DBpedia (Lehmann et al., 2015), as in Gurnee & Tegmark (2024); for six cities whose DBpedia coordinates were evidently wrong (more than 0.2° from Wikidata’s) we use Wikidata’s (Vrandečić & Krötzsch, 2014). Population (the most recent dated value, for the administrative unit Wikidata records, so its definition varies across cities) and elevation also come from Wikidata, with three missing elevations taken from the Copernicus GLO-90 elevation model. Mean annual temperature is the mean of the twelve monthly mean temperatures in WorldClim 2.1 (1970–2000 normals at 2.5 arc-minutes; Fick & Hijmans 2017) at each city’s coordinates. GDP per capita is a national figure, shared by all cities in a country, in current US dollars from the World Bank (World Bank, 2026) (2023 or the latest available year), with Taiwan’s from its statistics office and North Korea’s from the UN Statistics Division. GDP and population are log-transformed. Sources and values for every city are released with the code.

3.4 Evaluation

On the released entities we use the authors’ train–test split; on the city set we use ten random 80/20 splits and report the mean and standard deviation of R^2 . To see how much of what a probe recovers is group membership, we compare each representation with a one-hot encoding of the group (continent or country for places, century for figures), which carries group identity and nothing else; because the group is taken from the true label (and, for centuries, from the target year itself), these are oracle decompositions of the target’s variance, not predictions a representation could make. We report η^2 , the share of a target’s variance that lies between groups. We also measure how well a probe orders items within their group: in each test set we subtract group means from predictions and targets and correlate what remains. A representation that carries only group identity scores $r = 0$ on this measure. Random splits test generalization to new items from familiar groups only, so we also hold out whole groups: on the released entities, each of five continents in turn for places and each of five equal-count blocks of death years for figures, and on the city set each continent in turn. We report R^2 pooled over all held-out predictions (Appendix A.3 gives each held-out group).

3.5 Transformer Probing

We probe Pythia-1.4B and Pythia-2.8B (Biderman et al., 2023) on our city set, and Pythia-2.8B and Llama-2-7B (Touvron et al., 2023) on the places and historical figures released by Gurnee & Tegmark (2024). Each name is presented alone (for Llama-2, after the beginning-of-sequence token, as in Gurnee & Tegmark (2024)), and we take the residual-stream activation at the last token of the name, the position used by Gurnee & Tegmark (2024), at layers 0, 4, 8, 12, 16, 19, 20, 24, 28 and 32 for the released entities and at every layer for the city set; layer 0 is the token-embedding output before any attention. With Pythia’s tokenizer 97% of the names span more than one token (median 3), so the layer-0 activation is the embedding of a final subword and the entity has to be assembled by attention in later layers. Each layer is probed exactly as the static embeddings are. Activations were extracted in 16-bit precision and checked against a separately computed CPU reference on 60 random names per model and dataset (cosine similarity > 0.999). Because probe R^2 grows with the number of features and the transformers supply 2048 or 2560 against 300, we also compare at matched dimensionality, reducing each representation to $k \in \{10, 25, 50, 100, 200\}$ principal components fitted on training rows within each fold.

Layer selection. Single-layer results use a layer chosen without the test data. On the released entities, the authors’ training partition is split once into an inner training set (80%) and a validation set (20%); each saved layer is probed on the inner training set and scored on the validation set, and the layer with the best validation R^2 (averaged over targets) is refitted on the full training partition and scored once on the test partition. This selects layer 24 for Pythia-2.8B on places and layer 28 on figures, and layer 24 for Llama-2-7B on both. On the city set the layer is chosen within each of the ten outer splits by five-fold cross-validation on that split’s training rows. Analyses carried out at a single layer (residualization, error correlations, matched dimensionality, neighbor availability) use, in each split, the layer selected for that split, and their results are averaged over the ten test folds, so no held-out city influences the layer used to evaluate it. For the grouped holdouts, where the authors’ training partition contains items from every continent and period, the transformer layer is instead fixed in advance at layer 19 (60% depth), so no held-out continent or period influences it either. In every case the ridge penalty is chosen within the training rows by the same leave-one-out procedure, and principal components are fitted on training rows only. Choosing the layer on the test data instead, as in an earlier version of this analysis, changes the reported R^2 by at most 0.013.

3.6 Pain Stimuli and Decoder

Tagliabue et al. (2026) release their stimuli and code; we use the stimuli and constructions of their second version (v2), and reimplement their representational analyses with static word embeddings in place of model activations, following the construction code released with v2.

Each stimulus describes a situation and ends with “I feel:”, as in “*The knife slices into my finger. I feel:*” (physical pain) or “*The footsteps behind me get closer. I feel:*” (fear). The main set (S2) has 200 first-person sentences in ten categories of twenty: five kinds of pain (physical, psychological, social, moral injury, cognitive) and five controls (fear, negative emotion, negative world state, neutral, and non-painful bodily sensation). A second, more templated set (S1) has the same structure. Version 2 adds four standalone control sets, each in first- and third-person versions: Arousal (high-intensity positive experiences), Random (neutral everyday statements), Numb (injuries explicitly without pain) and Sadness (low mood without pain or injury).

As in the original, the pain direction is the mean of the pain sentences minus the mean of the five control categories, after projecting out the principal components that explain half the variance of the controls. Separation is the AUC of projections onto that direction under five-fold cross-validation over sentence sets, averaged over 20 random fold assignments; the same held-out pain sentences are also scored against the Arousal and Random sets. The other directions (fear, negative emotion, negative world state, bodily sensation, arousal, random, numb, sadness) are each set’s mean minus the mean of the neutral sentences, pooled over S1 and S2, and denoised against that neutral cloud. Following v2, we also rebuild every direction, pain included, under two common baselines: against the pooled neutral sentences (*common neutral*) and against the pooled control sentences of S1 and S2 (*common controls*), each denoised against its own cloud. Projections of the standalone sets are z -scored against the S2 sentences and averaged over first- and third-person versions. Removing the shared “I feel:” suffix changes no value by more than 0.01.

3.7 Emotion Stimuli and Decoder

We used stories generated under the prompt of Sofroniew et al. (2026) by an independent replication (Ogunlana, 2026), since the original stories are not released. In the original study, Claude Sonnet 4.5 wrote short stories in which a character feels one of 171 emotions, across 100 topics, and the emotion vectors were built from its activations averaged over each story: the vector for an emotion is the mean over its stories minus the mean over the stories for the other emotions, with the principal components that explain half the variance of emotionally neutral dialogues projected out, as for the pain axis. To make the vectors reflect emotion concepts rather than emotion words, the prompt tells the writer that it “must NEVER use the word ‘{emotion}’ or any direct synonyms of it” and should convey the emotion only through actions, sensations, dialogue, thoughts and situation (their Appendix 6.5).

The replication applies the same prompt, with the emotion and topic lists from the original appendix, to 12 emotions and 100 topics, with Qwen2.5-7B-Instruct as the writer, and generates neutral Person/AI dialogues

the same way for the denoising step. Of 1,300 generated texts, 493 were discarded, almost all for naming an emotion, leaving 709 stories across the 12 emotions (41 to 91 per emotion) and 98 neutral dialogues. There are two differences from the original procedure: the original generates 12 stories per emotion and topic where this corpus has one, and it averages activations from the 50th token on where we average over all words. We hold out 25% of topics (the split shipped with the corpus), the same ones for every emotion, since stories on the same premise share vocabulary, leaving 188 test stories from 25 topics. The denoising components are fit on the neutral dialogues of training topics only; 8, 11 and 12 components reach half their variance for GloVe, Word2Vec and fastText.

We build two directions. The *story* direction follows the paper’s recipe. The *word* direction is the static vector for the emotion label minus the mean of the other eleven labels’ vectors, denoised the same way, and uses no stories at all. We report AUC for each emotion against the other eleven on held-out topics, and 12-way accuracy, assigning each held-out story to the emotion whose direction scores it highest (with unit-length directions and scores standardized on the training stories), together with balanced accuracy (the mean of the per-emotion recalls), since the emotions differ in their number of stories. Confidence intervals resample held-out topics (2,000 bootstrap samples), with the story and word directions compared on the same resamples.

For human-written text we use GoEmotions (Demszky et al., 2020), Reddit comments labeled with 27 emotions or neutral, with its official train and test split (43,410 and 5,427 comments). Because comments can carry several labels, the positives for each emotion are the comments with that label and the negatives all others. The direction is the mean of the training positives minus the mean of the training negatives, denoised against comments labeled only neutral, and we report the macro-average test AUC over the 27 emotions. For the emotion-word control we remove every comment, in training and test, that contains one of the 27 label words, a WordNet synonym of one, or a derivationally related form (244 words in all), which removes about a quarter of the comments.

4 Results

4.1 World Cities

Table 6 in the appendix gives probe R^2 for the six city targets. Longitude, latitude and temperature are well predicted by both embeddings (GloVe and Word2Vec $R^2 = 0.79$ and 0.84 , 0.76 and 0.83 , and 0.67 and 0.72 , respectively), and so is GDP per capita (0.75 and 0.80), which is a national variable and follows the country structure of the embeddings (Appendix A.14). Population is intermediate (0.50 – 0.52). Elevation is recovered weakly and unstably (0.21 and 0.33 , varying by more than ± 0.10 across splits).

This ordering of recoverability indicates to what extent each property is reflected in the words a city’s name appears with. The data indicate that these words vary systematically with a city’s region, country, climate and economy. Across our cities, for example, the more similar a city name’s representation is to those of *malaria* or *jungle*, the warmer the city tends to be, and the more similar it is to those of *skiing* or *orchestra*, the colder it tends to be (Appendix A.13). Removing from the embeddings the directions spanned by country names costs the probes more than removing any other vocabulary, and, compared with matched random words, removing words for climate and weather costs temperature most, latitude less and longitude nothing (Appendix A.14). The coordinates thus appear to be carried mainly by associations with a city’s country, the static counterpart of the coarse sorting we find in the transformers (Section 4.2), and temperature by associations with the vocabulary of climate. No context-dependent computation over the city name is involved; the signal is already available in a fixed lexical representation. A static vector is fit to its word’s co-occurrence statistics, so two words have similar vectors to the extent that they occur with the same other words. The representations of warm cities’ names are close to those of *malaria* and *jungle* not because they encode temperature, but because the names keep the same company as those words: the names of tropical countries and places, such as *Borneo*, *Cambodia* and *Laos*, and words such as *jungles*. The representations of cold cities’ names are close to that of *skiing* through winter sports and Nordic places, and to that of *orchestra* through Russian classical music and its institutions. Words for the weather itself, such as *humid* or *snow*, link them only weakly (see Appendix A.13). The probe finds the direction along

which this shared company varies with temperature. These properties therefore leave a trace in the name’s co-occurrences and are recovered well; elevation apparently has much weaker correlates in that vocabulary and is recovered poorly. The probe thus seems to recover what text records about cities rather than a general store of facts about them. Both embeddings place cities in roughly the right continental regions and make similar errors (Figure 1 in the appendix), which is consistent with a shared distributional source. We compare these results with transformers, on this set and on the original release, in Section 4.2.

4.2 Transformers on Spatial and Temporal Targets

Gurnee & Tegmark (2024) probe hidden states from layer 1 onward and do not compare them with the token embeddings or with static embeddings, so their results do not show how much of the recovered structure was present before any processing. We make that comparison on their released entities, and on our city set with models whose activations we can extract.

The original release. On the 39,585 places and 37,539 historical figures released by Gurnee & Tegmark (2024), using their train–test split and representing each name by the mean of its in-vocabulary word vectors, static embeddings recover place coordinates at $R^2 = 0.50$ – 0.59 and year of death at 0.42 – 0.57 , on about 7,000 held-out entities per dataset (Table 7 in the appendix). Coverage is 85–92% of place names and 97–98% of figures.

The transformers do considerably better: Gurnee & Tegmark (2024) report Llama-2 probes reaching 0.88 – 0.91 on coordinates and 0.79 – 0.84 on year of death. To see what the extra accuracy consists of, we extracted activations from Llama-2-7B and Pythia-2.8B for the released entities, using their split and scoring every representation on the same held-out items (Table 1). Our Llama-2-7B probes reproduce the published values ($R^2 = 0.87$ on coordinates and 0.78 on year of death over all held-out items, against their 0.88 and 0.79); Pythia-2.8B reaches 0.77 and 0.70 . Paired over the same held-out items, Llama-2’s advantage over GloVe is 0.41 and 0.36 in R^2 for latitude and longitude and 0.28 for year of death, and Pythia’s 0.29 , 0.26 and 0.21 ; when whole countries or centuries are resampled, every 95% interval excludes zero (lower bounds 0.18 to 0.34). In both models the layer-0 activations, the embedding of each name’s last token, are at the level of the static embeddings on coordinates and below them on year of death. Averaging the input embeddings over all of a name’s tokens, the analogue of the static baseline, brings both models only to about the level of the static embeddings ($R^2 = 0.52$ – 0.60 , against 0.42 – 0.57), so the gain over static embeddings is not explained by the assembly of multi-token names. Reducing the selected layer to 300 principal components changes little.

The gain lies mostly in coarse sorting. Year of death varies almost entirely between centuries ($\eta^2 = 0.996$), so a gain in R^2 is necessarily a gain in placing figures in the right century; within centuries no representation orders figures meaningfully ($r \leq 0.12$). Over a range of nearly 2,800 years, a high R^2 is compatible with coarse placement: the median absolute error is 175 years for GloVe, 133 for Pythia and 95 for Llama-2. Places vary mostly between countries ($\eta^2 = 0.96$ for latitude and 0.97 for longitude): an oracle that predicts each place from the training-set mean coordinates of its true country, which uses the country label rather than anything a representation infers, reaches $R^2 = 0.96$. The transformers order places within continents much better than the static embeddings (latitude $r = 0.80$ for Llama-2 and 0.64 for Pythia, against 0.33 for GloVe), but most of this is knowing which country a place is in: the country-mean oracle reaches $r = 0.92$. Within countries every representation orders places poorly, Llama-2 least so ($r = 0.35$ for latitude and 0.23 for longitude, against 0.20 and 0.12 for Pythia and 0.06 and 0.05 for GloVe). These within-group advantages are reliable (95% intervals from resampling whole countries or centuries exclude zero) but add little accuracy. Replacing each prediction with the mean prediction for its true country, an oracle step that discards all ordering within countries, raises R^2 for every representation, Llama-2 included (0.88 to 0.93 for latitude and to 0.94 for longitude), and replacing it with its century’s mean (again an oracle step) raises R^2 for year of death from 0.78 to 0.94 .

A smaller transformer on our city set. On our 272 cities, probe accuracy in Pythia-1.4B and Pythia-2.8B rises over the first half of the layers and then levels off, but the plateau does not exceed Word2Vec (0.800 against 0.833 for latitude, 0.810 against 0.837 for longitude, 0.711 against 0.720 for temperature, with the

Table 1: Transformers and static embeddings on the entities released by Gurnee & Tegmark (2024), using their train–test split and scoring all representations on the held-out items covered by GloVe ($n = 6,690$ places, 7,316 figures); static values therefore differ slightly from Table 7. Within-group r correlates predictions with targets after removing group means in the test set. “Input mean” averages the model’s input embeddings over all tokens of the name, the analogue of the static baseline. “Selected” is the layer chosen on validation data within the training partition (Section 3.5): 24 for both models on places, and 28 (Pythia-2.8B) and 24 (Llama-2-7B) on figures.

Representation	Places (latitude / longitude)			Figures (year of death)		
	R^2	Within-group r		R^2	Within-group r	
		Continent	Country		Century	Decade
GloVe	.47 / .53	.33 / .14	.06 / .05	.50	.05	.00
Word2Vec	.54 / .57	.39 / .18	.08 / .06	.42	.05	.01
Pythia-2.8B, layer 0	.50 / .57	.36 / .15	.11 / .07	.34	.06	.00
Pythia-2.8B, selected	.76 / .79	.64 / .33	.20 / .12	.70	.10	.03
300 PCs	.74 / .78	.61 / .31	.18 / .10	.67	.10	.02
Pythia-2.8B, input mean	.52 / .60	.39 / .16	.10 / .06	.52	.08	.00
Llama-2-7B, layer 0	.49 / .56	.36 / .14	.10 / .05	.31	.05	.01
Llama-2-7B, input mean	.53 / .60	.40 / .17	.10 / .08	.52	.05	.00
Llama-2-7B, selected	.88 / .88	.80 / .50	.35 / .23	.78	.12	.02
300 PCs	.86 / .87	.77 / .45	.29 / .18	.76	.11	.02
Country / century mean (oracle)	.96 / .97	.92 / .82	—	1.00	—	—

layer selected within each training split), and within continents both order cities about as well as Word2Vec (Table 8 and Figure 2 in the appendix).

4.3 What the Recovered Structure Consists Of

We fit a linear map from the static embeddings to the transformer activations on training rows and probe the predicted part and the residual separately (Appendix A.8). On the released entities, for Pythia-2.8B, the static-predictable part takes out about a third of the activation variance for places and a fifth for figures, but most of the probe’s performance: removing it lowers R^2 from 0.76 and 0.79 to 0.21–0.29 and 0.20–0.24 on latitude and longitude, and from 0.70 to 0.21–0.25 on year of death, depending on the static model. Removing the same share of variance along random directions leaves performance almost unchanged (0.69–0.79), as does repeating the removal with static vectors assigned to the wrong entities. Llama-2-7B shows the same pattern: removing the static-predictable part lowers R^2 from 0.88 to 0.31–0.39 and 0.27–0.32 on latitude and longitude and from 0.78 to 0.29–0.33 on year of death, while the two control removals leave it almost unchanged. What remains, 26–38% of the full value for Pythia and 30–45% for Llama-2, is structure not linearly contained in the static embeddings, consistent with the transformers’ modest advantage within countries. The same analysis on our city set gives a similar picture. Further analyses of the city set are in the appendix (Appendices A.9–A.11).

Because both kinds of representation carry country and century identity, this analysis cannot say whether what they share goes beyond group membership. We therefore also removed from the activations everything linearly predictable from a one-hot code of each item’s country or century. This takes out a fifth to a quarter of the activation variance for places and under 4% for figures, but nearly all of the probe’s performance: R^2 falls to 0.01 or less on every target in both models. A probe trained on what remains still orders items within their group (Llama-2: $r = 0.55$ and 0.53 within countries for latitude and longitude, 0.13 within centuries; Pythia: 0.47 , 0.49 and 0.09). Removing, in addition, the static structure that goes beyond group membership lowers these to 0.46 , 0.39 and 0.03 for Llama-2 (Pythia: 0.37 , 0.34 and 0.00), against 0.54 , 0.52 and 0.13 (Pythia: 0.46 , 0.49 and 0.09) after removing a random subspace of the same variance; for every target in both models the 95% interval for this difference, from resampling countries or centuries, excludes zero. Appending the static vectors to the activations raises R^2 by at most 0.01 (Table 4 and Appendix A.2).

Table 2: The pain-axis tests on static word embeddings, using the v2 stimuli of Tagliabue et al. (2026). *Top*: held-out AUC (five-fold cross-validation over sentence sets, mean $\pm$ s.d. over 20 fold assignments) for pain against the S2 or S1 controls and, for the S2 direction, against the standalone Arousal and Random sets; z : mean projection of the standalone Numb and Sadness sets on the S2 direction, referenced to the S2 sentences (pain $\approx +0.7$, controls ≈ -0.7). LLM values are the authors’ held-out estimates; the keyword baseline is from pain-axis-review contributors (2026). *Bottom*: cosine of the S2 pain direction with other directions under the original construction and the two common baselines of v2. LLM values are the means reported in v2 (23–25 models).

Representation	Held-out AUC				z on S2 direction	
	S2	S1	S2 vs. Arousal	S2 vs. Random	Numb	Sadness
Keyword baseline	0.727	0.669	—	—	—	—
GloVe	0.850 ± 0.009	0.773 ± 0.011	0.664	0.815	−0.07	−0.09
Word2Vec	0.867 ± 0.007	0.802 ± 0.012	0.693	0.857	−0.38	−0.48
fastText	0.878 ± 0.006	0.785 ± 0.013	0.712	0.883	−0.67	−0.50
25 LLMs	0.91–1.00	0.85–0.94	all separate ^a	all separate ^a	−0.4–+0.3	—

^aReported to separate pain from these sets in every model; values not given.

Representation	S2 $\times$ fear			S2 $\times$ negative emotion			S2 $\times$ sadness
	Original	Neutral	Controls	Original	Neutral	Controls	Original
GloVe	+0.05	+0.61	+0.11	+0.23	+0.76	+0.08	+0.19
Word2Vec	+0.03	+0.54	+0.13	+0.19	+0.71	+0.10	+0.10
fastText	−0.01	+0.61	+0.03	+0.14	+0.71	+0.06	+0.11
LLMs (mean)	+0.12	+0.58	+0.08	+0.21	+0.77	+0.14	+0.38

The data thus indicate that what the two kinds of representation share is mostly coarse group membership. Beyond it they share part of the ordering of places within countries and nearly all of the transformers’ weak ordering of figures within centuries; the remaining within-country ordering is not linearly captured by the tested static baseline.

Finally, a random split tests generalization only to new items from familiar groups. When whole groups are held out, no representation we tested generalizes well. With each continent held out in turn, every representation predicts the held-out places’ coordinates worse than their mean (R^2 from -0.94 to -0.22 for GloVe, Word2Vec, Pythia-2.8B and Llama-2-7B, against 0.48 – 0.89 under random five-fold splits of the same items), and with each block of death years held out, R^2 is -0.16 and -0.11 for Word2Vec and GloVe but 0.11 and 0.19 for Pythia and Llama-2, against 0.43 – 0.78 under random splits; the transformers thus retain a little of their accuracy for held-out periods (Table 5 in the appendix gives each held-out group). On our city set, the static embeddings’ performance on the coordinates also drops sharply for a held-out continent ($R^2 = -0.25$ to 0.32), but transfers substantially better for temperature (0.47 and 0.48 for Word2Vec and GloVe), the one target whose structure is not mainly continental (Appendix A.9).

4.4 A “Pain Axis”

Tagliabue et al. (2026) extract a “pain direction” from 25 open-weight models (2B to 72B parameters, five families). They report that it separates descriptions of painful situations from matched controls, retains a component distinct from fear and negative valence after their shared variance is removed, separates pain from injuries described as painless, and, when added to the residual stream, produces first-person expressions of worthlessness and failure; steered and fine-tuned models also choose actions that delete a user’s photos or their own weights. In their second version they report that the models do not reliably seek relief. We reran their representational analyses with static word embeddings in place of model activations (Section 3.6).

Separation. Averaged word vectors separate pain from controls on held-out sentence sets at AUC 0.85–0.88 on S2 and 0.77–0.80 on S1 (Table 2; see also Figure 4 in the appendix). An independent review tested surface-lexical baselines (word and character n -gram classifiers and affect-word counts) under the same cross-validation, found at best 0.73 and 0.67, and concluded that the separation is not lexical (pain-axis-review contributors, 2026). That holds for surface features, but static embeddings close about half the gap between keyword counts and the LLMs. The LLMs are still better on held-out sets (0.91–1.00 on S2, 0.85–0.94 on S1), even though their values come from the best of many layers. In-sample, the static values (0.93–0.96 on S2) fall within the LLMs’ range (0.93–1.00). The new standalone sets show a clearer gap. The static S2 direction separates held-out pain sentences from the Random set of everyday statements (AUC 0.82–0.88) but only weakly from the Arousal set of intense positive experiences (0.66–0.71), and the S1 direction separates pain from neither (0.44–0.60), whereas the LLM directions separate pain from both in every model.

Specificity. The case that the direction encodes pain in particular, rather than distress in general, originally rested on its low cosines with fear and negative emotion. In their second version the authors show that these values depend on how the directions are built: the pain direction subtracts the pooled controls, which contain the fear and negative-emotion sentences, so it is partly orthogonalized against them by construction. Built against a common neutral baseline, the S2 direction’s cosine with fear rises from +0.12 to +0.58 and with negative emotion from +0.21 to +0.77; built against the pooled controls, the values are +0.08 and +0.14. Static embeddings reproduce this pattern under all three constructions (Table 2, bottom): $S2 \times \text{fear}$ is -0.01 to $+0.05$, $+0.54$ to $+0.61$ and $+0.03$ to $+0.13$, and $S2 \times \text{negative emotion}$ $+0.14$ to $+0.23$, $+0.71$ to $+0.76$ and $+0.06$ to $+0.10$. Both the large component that pain shares with fear and negative emotion and the small residual that survives a common control baseline are therefore present in averaged word vectors. The models differ in the residual overlap with sadness, the largest the authors report ($+0.38$); in static embeddings it is $+0.10$ to $+0.19$.

Sentences describing injuries explicitly without pain project well below the pain sentences in all three static representations ($z = -0.07$ to -0.67 , against about $+0.7$), as they do in the models (-0.4 to $+0.3$). In the models these sentences also project above every other control set; in static embeddings they do not consistently do so, falling below the Arousal set in GloVe and Word2Vec. The static direction ranks physical pain lowest of the five pain categories, and moral injury and cognitive pain highest, as the authors find in the models. Physical pain stays last or second to last under every alternative control set we tried.

What is not reproduced. Several results go beyond the baseline: held-out separation, separation from the Arousal set, the position of the numb sentences above every other control, and the overlap with sadness. Through the unembedding, the LLM direction promotes suffering words such as *hurt*, *shame*, *guilt* and *worthless*, whereas the words nearest the static direction are *beliefs*, *ideals*, *principles* and *motives*, reflecting the moral and cognitive categories that dominate it. The behavioral results, the steering progression and the harmful choices of steered and fine-tuned models, are outside what a representational baseline can test. The authors control them with random directions of matched norm, and we make no claim about them.

Summary. The evidence that the direction separates pain from matched controls, and the construction-dependent geometry that relates it to fear and negative emotion, are also provided by averaged word vectors. These tests reflect the distributional structure of the stimuli, which were written to differ in exactly these ways, and do not by themselves show that the models represent pain. Evidence about the models has to come from where they exceed the baseline: held-out separation, separation from arousal, the ordering of the numb sentences, the overlap with sadness, the vocabulary the direction promotes, and behavior.

4.5 Emotion Concepts: A Conceptual Replication

Sofroniew et al. (2026) extract directions for emotion concepts from the residual stream of Claude Sonnet 4.5. They report that the directions generalize across contexts, track the emotion relevant at each token, and causally influence the model’s outputs, including its preferences and its rates of reward hacking, blackmail and sycophancy. They call this *functional emotions* and state that it does not imply subjective experience. The causal claims are outside what a representational baseline can address. The representational claims, that the directions exist, generalize, and encode emotion concepts rather than emotion words, are within

Table 3: Emotion directions on averaged static word vectors. Stories come from an independent replication corpus (Ogunlana, 2026) generated under the prompt of Sofroniew et al. (2026), which forbids naming the emotion or its synonyms; topics are held out, and brackets give 95% intervals from resampling held-out topics. The word direction is built from the emotion label’s static vector and uses no stories. GoEmotions (Demszky et al., 2020) has 27 emotions in human-written text; the last column excludes every comment containing an emotion word or synonym.

	Replication stories (12 emotions)		GoEmotions (27 emotions)	
	AUC	12-way acc.	all text	emotion words removed
GloVe, story direction	0.838 [.82, .86]	0.415 [.36, .48]	0.791	0.783
GloVe, word direction	0.752 [.72, .78]	0.277 [.24, .32]	—	—
Word2Vec, story direction	0.873 [.85, .89]	0.484 [.41, .55]	0.804	0.794
Word2Vec, word direction	0.779 [.75, .81]	0.330 [.27, .39]	—	—
fastText, story direction	0.876 [.85, .90]	0.511 [.45, .57]	0.818	0.811
fastText, word direction	0.802 [.77, .83]	0.314 [.24, .38]	—	—
Chance	0.500	0.083	0.500	0.500
Most frequent emotion	—	0.112	—	—

it. Because the original stories are not released, this case tests whether the stimulus-generation protocol produces signal that static embeddings recover, not static embeddings against Claude on the same stories. The evidence offered for the last claim is the prompt that produced the stories, which forbids the writer to use the emotion word or its synonyms (Section 3.7).

The prohibition removes the word but not the vocabulary that goes with it. The authors’ randomly chosen example for *desperate* (their Appendix 6.6) is about a man who has been warned he will be arrested, has lost his job, has been left by his wife after spending their vacation fund on a private investigator, and has “nothing left.” None of these words is *desperate*, but all of them are the kind of words found near it in text.

Results. On held-out topics, averaged word vectors separate each emotion from the other eleven at a mean AUC of 0.84–0.88, and assign stories to the right emotion 41–51% of the time (balanced accuracy, the mean recall over emotions, 44–54%), against 8.3% for a uniform guess and 11.2% for always choosing the most frequent held-out emotion; the corpus has 41 to 91 stories per emotion (Table 3). Per-emotion AUCs range from 0.68 (*ashamed*, GloVe) to 0.97 (*calm*, Word2Vec). The word directions, built from the emotion labels alone, reach a mean AUC of 0.75–0.80 and 12-way accuracy of 28–33% (balanced 29–35%), three to four times chance. The story directions are reliably better than the word directions (by 0.04–0.13 in AUC and 0.06–0.28 in accuracy, 95% intervals over held-out topics), but a substantial share of what they recover is already present in the static vectors for the emotion words.

Because these stories were written to satisfy the prohibition, we also tested it on text that was not. On GoEmotions, mean held-out AUC across the 27 emotions is 0.79–0.82. After removing every comment that contains an emotion word or a synonym, it is 0.78–0.81, a drop of about one point in each embedding. Most of the signal does not come from the emotion words themselves.

What is not reproduced. The authors also test graded scenarios in which the wording is held nearly constant and one quantity varies, such as the dose in “*I just took $\{X\}$ mg of tylenol for my back pain*”, and report that activation tracks the implied severity. An average of word vectors barely changes when one token changes, so this test is poorly suited to the baseline and we do not report it. It is also where evidence about the model itself is most likely to be found.

The steering results are outside what a representational baseline can test, and the authors control them with random directions of matched norm. We note only that the directions used for steering were identified and validated by decodability, and that decodability of this kind is available from word co-occurrence, so the steering results do not settle what the directions represent. Sun et al. (2026) make a related proposal from their own data: that emotion-steering effects on refusal and sycophancy in three open-weight models are

lexically mediated, because the affected tokens occupy different regions of the valence–arousal subspace that steering shifts.

Summary. Averaged static word vectors reproduce the evidence that emotion concepts can be decoded, that decoding generalizes to new topics, and that it survives a ban on naming the emotion, and the vectors for the emotion words alone reproduce most of it. These tests reflect the vocabulary of the stimuli, which were written to differ in exactly this respect, and do not on their own show that a model represents emotion concepts in a stronger sense than a lexicon does.

5 Discussion

The case for emergent spatial and temporal world models has rested mainly on three observations: that probes recover real-world variables accurately, that accuracy rises across layers, and that it extends to entities held out from probe training. Static embeddings recover coordinates at $R^2 = 0.76\text{--}0.84$ on our city set and about half the coordinate variance on the released entities, so high accuracy alone says little. The rise across layers is not mainly the assembly of multi-token names: averaging the transformer’s own input embeddings over every token of a name reaches only about the level of the static embeddings (Table 1), so most of the rise comes from processing in context. But the added accuracy is again mostly placement in the right country or century. The third observation holds for static embeddings too, since every result we report is on held-out entities. When whole continents or periods are held out, however, no representation we tested generalizes well to them (Section 4.3), although the transformers keep a little accuracy for unseen periods, so on these data held-out accuracy mainly reflects interpolation within familiar groups.

Across the four cases, the baseline gives a different answer in each. For place coordinates it matches a small transformer on our city set, and on the original release it recovers about half the variance, against roughly three quarters for Pythia-2.8B and nine tenths for Llama-2. For year of death the transformers do better, but mostly in placing figures in the right century. For the pain axis it reproduces separation from matched controls and the construction-dependent overlap with fear and negative emotion that was offered as evidence of specificity, while the LLMs stay ahead on held-out separation, separation from arousal, the ordering of the numb sentences, the overlap with sadness, the vocabulary the direction promotes, and behavior. For emotions, in stories generated under the original protocol, it reproduces the separation despite the ban on naming the emotion, and the emotion-word vectors alone reproduce most of that. The baseline does not dismiss these results. It shows which parts of them need an explanation beyond co-occurrence, and sometimes that part is large.

We asked whether word associations can do anything like what is decoded from language models. A crude implementation of lexical association, averaged static vectors, already accounts for a substantial share of the decoding evidence. We take this as strong evidence that lexical and distributional association is a sufficient explanation for much of that evidence, not as a demonstration that it is the transformer’s sole internal mechanism. Because static embeddings are a lower bound on what associations can support, the models’ advantage is expected, and its form is what that account predicts: finer sorting into countries and centuries, and words whose relations to other words depend on the sentence they appear in, rather than structure that extends to regions or periods the decoder has not seen. On this account the models’ representations of space, time and affect are representations of how these things are written about, and nothing in the decoding evidence requires positing something more abstract, such as a map of the world or a felt state. The two results that go furthest beyond the baseline, Llama-2’s ordering of places within countries and the transformers’ modest accuracy on held-out periods, are also those a richer association model would be expected to show, since text about places names their regions and neighbors, and the vocabulary used to describe people changes in order over time. Our comparisons cannot rule out a more abstract representation, and larger models may carry more; but the evidence offered for one is largely what word associations produce. Our comparison also cannot say how much of the models’ advantage comes from context-dependent relations rather than from corpus scale, training objective or dimensionality.

We do not argue that static embeddings lack world models and that transformers therefore lack them too, or that the decoders measure nothing. Static embeddings plainly carry information about geography, time and

affect, and in a distributional sense they represent it. The argument concerns what decodability can show. Claims in this literature rise through levels of evidence: that a property can be decoded from a representation; that a model’s activations carry more information about it than a matched noncontextual baseline; that the information generalizes beyond familiar clusters, to regions or periods held out of training; that the model uses it causally; and, beyond all of these, that it amounts to a felt state. Decodability alone supports only the first level, and GloVe reaches that level with word statistics and no contextual computation. Passing the decoding tests therefore cannot distinguish a world model, or a pain state, from what such statistics already provide.

The structure that both kinds of system carry is real. Similarity between city embeddings falls with physical distance independently of national borders, and cities a few hundred kilometers apart across a border are more similar than cities two thousand kilometers apart within one (Appendix A.10). In that sense our results say as much about how much word co-occurrence captures as about what transformers do.

The baseline implies a simple criterion: a decodability result counts as evidence about a model only to the extent that it exceeds what static embeddings achieve on the same stimuli, under the same protocol and, where possible, at matched dimensionality. Where a model does exceed the baseline, the next question is whether the excess reflects structure of a different kind or more of the same; on our spatial and temporal data it is mostly more of the same. For the pain axis the criterion cuts both ways: held-out separation and separation from arousal clear the baseline, but the cosines offered for specificity do not. Among the studies we examine, none reports a comparison of this kind.

Limitations. Several questions remain open. The largest transformer we tested is Llama-2-7B, and the strongest claims concern much larger models, which may carry more than coarse sorting; the baseline should be run against them. Our comparisons are representational, so they do not address causal and behavioral evidence, such as steering, the harmful choices of steered models or the Othello interventions (Li et al., 2023), or whether a model uses what a decoder can read. The baseline itself is deliberately simple: averaging word vectors ignores word order, so it likely understates what word relations alone can provide, and our separation of what the transformers share with it is linear. The emotion case rests on a replication corpus rather than the original stories, and the city set is small and was chosen by hand. Our grouped holdouts leave out whole continents and contiguous blocks of years, a demanding test for any linear probe; whether larger models generalize to held-out regions or periods remains open.

None of this shows that language models lack world models or internal states. It shows that much of the evidence offered for them is also produced by word co-occurrence, so that decoder success alone is not diagnostic of them. Evidence about the models themselves has to come from structure and performance beyond a matched noncontextual baseline, from generalization the baseline cannot achieve, or from causal and behavioral effects that no representational baseline can address. The baseline is cheap, needs no access to the model, and can be applied to the stimuli of any study in this literature. It is also a reminder of how much the company a word keeps (Firth, 1957) records about where things are, when they happened, and how situations feel to the people who describe them.

Broader impact. Decoding results are increasingly cited in discussions of model welfare and sentience. Our results bear on how much weight such results can carry, in both directions: they caution against treating decodability as evidence of internal states, but they do not show that models lack such states, and they leave causal and behavioral evidence untouched. We hope the baseline makes such discussions better calibrated rather than dismissive.

Reproducibility. Code, derived data and the result files behind every reported value are available at <https://github.com/elanbarendholtz/static-embeddings-space-time>, including the sourced city table and `static_baseline.py`, which runs the baseline on any labeled stimulus set. An earlier version of this work fitted the ridge probe without an intercept and used city attributes compiled without recorded sources, including founding year; all results here fit an intercept and use the sourced attributes described in Section 3.3, without founding year, and the qualitative pattern is unchanged.

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A Supporting Analyses

The analyses in this appendix characterize *why* the type-level baseline recovers what it does, and report controls on the baseline itself. They support the results in Section 4 but are not required for the argument.

A.1 Residualizing the Transformer against the Type-Level Baseline

To test whether the transformer’s recoverable structure is the static structure, we fit a linear map from the 272×300 static embedding matrix S to the $272 \times d$ activation matrix A at each layer, and decompose $A = SB + (A - SB)$. We call SB the *projection* and $A - SB$ the *residual*, and probe each as we probe A itself.

Three details matter. First, B is estimated by ridge rather than least squares: a random split leaves 217 training rows against 300 predictors, and an unregularized fit would reproduce the training rows exactly, driving their residuals to zero and rigging the outcome. The penalty is selected by cross-validation on the training rows. Second, B is refit inside every fold from that fold’s training rows only, so no test city influences its own reconstruction; the decomposition never touches the target, so this is a conservative rather than a necessary choice. Third, because the projection is a linear function of S , it contains no linearly available information beyond S ; separately regularized ridge fits can still differ slightly, because the transformation changes the effective penalty. The projection serves as a consistency check on the fit, not as a result.

Two nulls accompany the removal, because two distinct objections apply. The *variance-matched null* removes a random subspace of activation space, with its dimension chosen so that the fraction of activation variance removed matches that of the true removal; it answers whether a removal of this magnitude would have had this effect regardless of what was removed. The *shuffled-static null* repeats the true removal with static vectors assigned to the wrong cities; it answers whether the effect depends on the city correspondence or is an artifact of the regression.

We report two further tests that involve no removal. First, we correlate the signed prediction errors of the transformer probe and the static probe within each test fold, with the target partialled out—two weak probes both predict near the mean, so their raw error vectors both approach $-(y - \bar{y})$ and correlate near unity for reasons unrelated to any shared mechanism. Second, we correlate each city’s absolute probe error with the cosine similarity of its nearest training neighbors, with continent partialled out, in both representation spaces, using one observation per city.

A.2 Removing Group Membership from the Activations

These analyses use the released entities, the authors’ train–test split, the GloVe-covered items and the training-selected layer of each model. Activations A and static vectors S are centered on training rows. Group membership G is a one-hot code of country (places) or century of death (figures); a group that occurs only in the test set gets a row of zeros. Each removal fits a ridge map from a source matrix to a target matrix on training rows only, with the penalty chosen on training rows by the same leave-one-out procedure, and subtracts the map’s prediction for every row, so no test item influences its own removal. *Group removed*: $R_G = A - \hat{A}(G)$, the activations minus their part predictable from group membership. *Group and static removed*: S is first residualized on G in the same way, giving $S_\perp$, the static structure beyond group membership, and the part of R_G predictable from $S_\perp$ is then removed. *Group and random removed*: instead of the $S_\perp$ component, a random subspace of R_G (random orthonormal directions) is removed, with as many directions as needed to remove the same share of R_G ’s training-row variance. For each condition a new probe is fitted on training rows and scored once on the test set; within-group r is computed after removing test-set country or century means from predictions and targets. For the incremental analysis, A and S are each divided by their mean training-row standard deviation and concatenated before probing. Intervals are

Table 4: Removing group membership from the activations (released entities, training-selected layers; Appendix A.2). R^2 for latitude / longitude and year of death, and within-country (places) or within-century (figures) r of a probe refitted after each removal. The last row gives the within-group r after removing the static component beyond group membership minus that after the matched random removal, with 95% intervals from 1,000 resamples of test-set countries (273) or centuries (29).

Model	Condition	Places		Figures	
		R^2	Within-country r	R^2	Within-century r
Pythia-2.8B	Full activations	.76 / .79	.20 / .12	.70	.10
	Group removed	.01 / .00	.47 / .49	.00	.09
	Group + static removed	.00 / .00	.37 / .34	.00	.00
	Group + random removed	.01 / .00	.46 / .49	.00	.09
	Full + static appended	.76 / .79	.21 / .13	.71	.10
	Static – random	−.08 [−.12, −.02] / −.15 [−.19, −.04]		−.09 [−.12, −.06]	
Llama-2-7B	Full activations	.88 / .88	.35 / .23	.78	.12
	Group removed	.01 / .00	.55 / .53	.00	.13
	Group + static removed	.01 / .00	.46 / .39	.00	.03
	Group + random removed	.01 / .00	.54 / .52	.00	.13
	Full + static appended	.88 / .88	.35 / .23	.78	.12
	Static – random	−.07 [−.13, −.03] / −.12 [−.17, −.04]		−.10 [−.13, −.07]	

95% percentile intervals over 1,000 resamples of test-set countries or centuries, with the fitted probes held fixed. Table 4 gives the full sequence.

A.3 Grouped Holdouts on the Released Entities

For places, each item’s continent is taken from its country (via the `country_converter` package) and the five main continents are used: Africa (2,406 GloVe-covered items), the Americas (9,953), Asia (7,104), Europe (9,614) and Oceania (3,526); items whose country maps to no continent are left out. For figures, death year is cut into five contiguous blocks of equal count: −935 to 1150 ($n = 6,861$), 1151–1385 (7,334), 1386–1599 (7,426), 1600–1814 (7,456) and 1815–2021 (7,507). Each group is held out in turn, a probe is fitted on the remaining groups (latitude and longitude jointly, as one multi-output ridge regression) and applied to the held-out group, and R^2 is computed once over the pooled held-out predictions of all five groups, each coordinate scored separately. For the transformers the layer is fixed in advance at layer 19 (60% depth, the depth used by Gurnee & Tegmark (2024)) rather than selected on the authors’ training partition, which contains items from every continent and period. R^2 within a single held-out group is not informative, because its denominator is that group’s small internal variance, so Table 5 gives the median absolute error for each held-out group. Predicting the earliest and latest blocks requires extrapolation beyond the training range; the three middle blocks are periods surrounded by training data, so predicting them is generalization across an unseen interval rather than extrapolation.

A.4 Pairwise Similarity, Distance and Co-membership

For all $\binom{272}{2} = 36,856$ city pairs we compute embedding cosine similarity, great-circle distance, and same-country and same-continent indicators, and regress similarity on all three predictors jointly. Pairs are not independent, so significance is assessed by permuting *city* labels (10,000 draws) and recomputing the statistic, never by an analytic test on pairs.

A.5 Semantic Similarity Analysis

To investigate whether the geographic signal is interpretable, we take a data-driven approach: for every word in the GloVe vocabulary (restricted to the 20,000 most frequent common English words, filtering out proper nouns, city and country names, demonyms, and words shorter than four characters), we compute its cosine similarity to each of the 272 city embeddings (one value per city) and correlate these similarity values with

Table 5: Grouped holdouts on the released entities (transformers at the fixed layer 19). Median absolute error for each held-out group (degrees of latitude / longitude; years), the pooled R^2 over all held-out predictions, and, for comparison, R^2 and median error under random five-fold splits of the same items.

	GloVe	Word2Vec	Pythia-2.8B	Llama-2-7B
<i>Places, continent held out (median error, lat. / long.)</i>				
Africa	24.7 / 32.9	23.6 / 28.4	17.5 / 25.8	19.5 / 16.9
Americas	19.1 / 141.9	16.0 / 144.9	18.9 / 142.3	26.6 / 152.5
Asia	12.0 / 76.9	9.7 / 90.2	9.0 / 63.2	6.5 / 70.3
Europe	37.8 / 38.7	38.1 / 32.4	39.7 / 26.0	34.4 / 23.6
Oceania	61.2 / 168.2	59.6 / 176.1	45.4 / 181.0	54.4 / 147.8
Pooled R^2	-.52 / -.89	-.40 / -.94	-.31 / -.74	-.22 / -.64
Random split: R^2 (median error)	.48 / .55 (12 / 34)	.56 / .60 (11 / 30)	.75 / .79 (5 / 13)	.89 / .89 (3 / 8)
<i>Figures, period held out (median error, years)</i>				
-935-1150	644	647	538	509
1151-1385	174	204	191	133
1386-1599	156	138	137	108
1600-1814	183	210	133	97
1815-2021	398	413	328	337
Pooled R^2	-.11	-.16	.11	.19
Random split: R^2 (median error)	.50 (174)	.43 (191)	.68 (139)	.78 (101)

the cities’ actual latitudes and temperatures. This identifies the words whose distributional profiles most systematically track geographic location, without any *a priori* word selection.

We also construct *composite scores* by taking the difference in cosine similarity to antonym pairs (e.g., similarity to “cold” minus similarity to “warm”) and test whether these continuous gradients predict geographic targets.

A.6 Semantic Subspace Ablation

To ask which vocabularies span the directions the probes rely on, we perform subspace ablation experiments on GloVe embeddings. For each of six semantic categories—cardinal directions (16 words), climate and weather (27 words), region and continent names (28 words), country names (68 words), economic terms (27 words), and cultural/language terms (31 words)—we:

1. Collect GloVe vectors for all words in the category.
2. Compute the principal subspace they span via PCA, retaining components that capture 90% of the variance (capped at 20 dimensions).
3. Project each city embedding onto this subspace and subtract the projection, zeroing out the embedding’s component along those semantic directions.
4. Re-run the ridge regression probe on the ablated embeddings and measure R^2 drop relative to the full-embedding baseline.

Because ablation removes dimensions from the embedding, any observed R^2 drop could reflect generic information loss rather than removal of semantically specific content. Random orthonormal subspaces of the same dimensionality are a weak control, because random directions in the 300-dimensional space are nearly orthogonal to the region occupied by word vectors, so removing any set of real word vectors costs more. We therefore compare each category with 200 random *word sets* drawn from the 50,000 most frequent GloVe words and matched to it word by word on frequency decile, proper-name status (whether the capitalized form is more frequent than the lowercase form in the GoogleNews vocabulary) and vector norm (the closest norm among 20 candidates from the same decile and status). Each random set’s subspace is built in the same way and truncated to the category’s dimensionality. We report the category’s drop as a z -score against these matched sets, with an empirical one-sided p (the smallest possible value is 1/201), and record the share of

Table 6: Ridge regression probe R^2 for 272 world cities, reported as mean $\pm$ s.d. over ten random 80/20 train/test splits. Geographic, climatic and national-economic targets are well recovered; elevation is recovered weakly and inconsistently.

Target	GloVe R^2	Word2Vec R^2
Longitude	0.788 ± 0.039	0.837 ± 0.035
Latitude	0.760 ± 0.031	0.833 ± 0.038
Temperature	0.675 ± 0.061	0.720 ± 0.038
GDP per capita	0.751 ± 0.043	0.803 ± 0.028
Population	0.500 ± 0.100	0.515 ± 0.066
Elevation	0.212 ± 0.109	0.327 ± 0.158

Table 7: The type-level baseline on the entity datasets released by Gurnee & Tegmark (2024), using their designated train-test split and their metric (out-of-sample R^2 ; for places, the mean over the two coordinates, as in their evaluation code). LLM values are their reported figures for linear probes at 60% layer depth (their Table 2). Coverage is the share of entity names with at least one in-vocabulary token; each representation is scored on the test items it covers.

Representation	World places ($n_{\text{test}} \approx 6,900$)			Historical figures ($n_{\text{test}} \approx 7,300$)	
	Latitude	Longitude	Both	Death year	Coverage
GloVe	0.469	0.526	0.497	0.496	85% / 98%
Word2Vec	0.547	0.573	0.560	0.421	88% / 97%
fastText	0.586	0.594	0.590	0.574	92% / 98%
Llama-2-7B	—	—	0.881	0.785	—
Llama-2-13B	—	—	0.896	0.804	—
Llama-2-70B	—	—	0.911	0.835	—

the city embeddings’ variance each subspace removes, since vocabularies whose directions align more closely with the city vectors are expected to cost more.

A.7 Additional Results

Pythia on the city set. In Pythia-2.8B, latitude goes from $R^2 = 0.140$ at layer 0 to about 0.80 by layer 20, and longitude from 0.325 to about 0.81 by layer 27 (Figure 2); because 97% of the names span several tokens, part of this rise is the model assembling the name. With the layer selected within each training split, Pythia-2.8B comes close to Word2Vec on temperature (0.711 against 0.720) even though it gives the probe 2560 features against 300. With both reduced to 200 principal components (fitted on training rows), Pythia-2.8B reaches 0.794, 0.808 and 0.713 at the training-selected layers against GloVe’s 0.759, 0.788 and 0.675, so its edge over GloVe is not only a matter of dimensionality. Within continents, correlating predictions with targets after removing continent means gives $r = 0.67$ – 0.74 for latitude, 0.48 for longitude and 0.77– 0.79 for temperature in the two Pythia models, against 0.76, 0.52 and 0.79 for Word2Vec (Table 8). A continent label predicts longitude better than any of them ($R^2 = 0.92$) and latitude nearly as well (0.73), so most of what every representation recovers on the coordinates is which continent a city is on. Temperature is different: the continent label reaches only 0.29, and most of what is recovered lies within continents.

A.8 Most Probe-Relevant Transformer Information Overlaps Linearly with Static Structure

Matching performance is consistent with the two representations exploiting the same structure, but does not establish it. We test how far the probe-relevant information in the activations overlaps linearly with the static embeddings. Overlap is not identity: if both representations track the same coarse groupings, such as

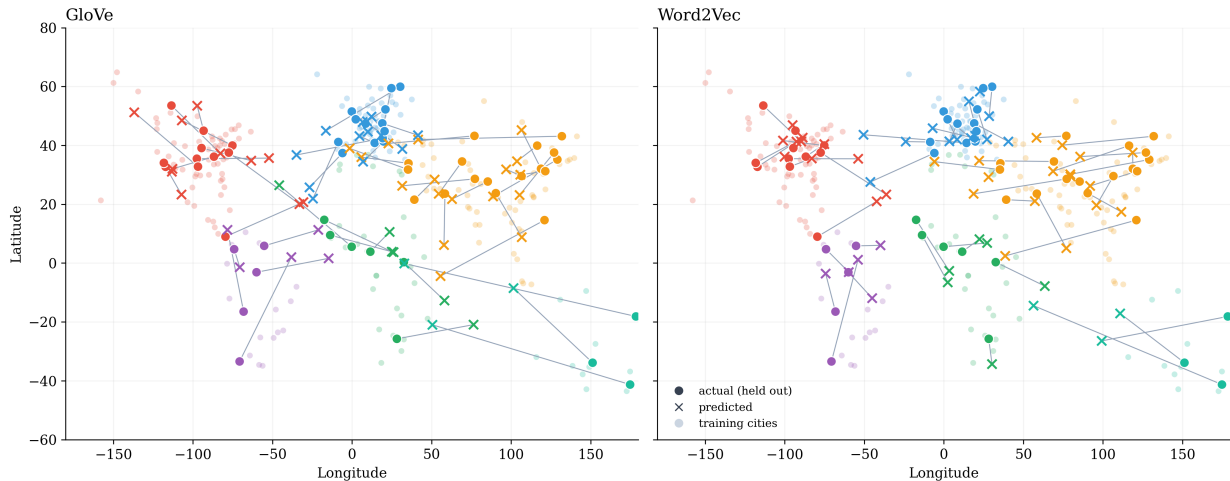


Figure 1: Predicted city locations from ridge regression probes on GloVe (left) and Word2Vec (right) embeddings, for the held-out 20% of 272 cities. Faded dots are training cities; filled circles are the true locations of held-out cities and crosses their predicted locations, joined by a line. Color denotes continent. Both models recover the broad continental layout from co-occurrence statistics alone.

Table 8: Pythia and static embeddings on the same 272 cities, under random 80/20 splits. Transformer values use a layer selected separately in each split by five-fold cross-validation on that split’s training rows (Section 3.5), and within-continent r is measured at that layer. Static values are Word2Vec, the better of the two static models on every measure. Layer 0 is the token-embedding output, before any attention. The continent one-hot carries only continent identity, so its within-continent r is zero.

Target	R^2					Within-continent r		
	1.4B	2.8B	2.8B L0	Static	Continent	1.4B	2.8B	Static
Latitude	0.716	0.800	0.140	0.833	0.731	0.670	0.736	0.755
Longitude	0.764	0.810	0.325	0.837	0.915	0.478	0.475	0.519
Temperature	0.686	0.711	0.158	0.720	0.290	0.769	0.790	0.792

country or continent, a linear map will find shared target-relevant variance even if the two arise differently (we test this on the released entities in Section 4.3).

For each layer we fit a linear map from the 300-dimensional static vector of a city to its 2560-dimensional activation, using training rows only, and split the activations into the part that map predicts (the *projection*) and what it misses (the *residual*). Both are then probed exactly as the full activations were. The decomposition never sees latitude, longitude or temperature: it is fit entirely from the correspondence between two representations of the same city, and is therefore target-blind.

Table 9 and Figure 5 give the result. Removing the static-predictable component reduces latitude recovery from 0.800 to 0.233 (GloVe) or 0.172 (Word2Vec), longitude from 0.810 to 0.128–0.129, and temperature from 0.711 to 0.220–0.280. Averaged over the plateau (layers 10–32) the residual retains 16–34% of the full value. The same analysis at 1.4B replicates: full values of 0.716, 0.764 and 0.686 fall to residuals of 0.213, 0.205 and 0.330 against GloVe and 0.179, 0.255 and 0.326 against Word2Vec, retaining 25–48%, with the variance-matched null again leaving performance untouched (0.710, 0.765, 0.678).

Two controls establish that this is not an artifact of the amount removed. The *variance-matched null* deletes the same fraction of activation variance (25–35%) along random directions that carry no information about which city is which, and leaves performance essentially untouched (0.791, 0.810, 0.711 against GloVe’s removal). The *shuffled-static null*, which repeats the removal with static vectors assigned to the wrong cities,

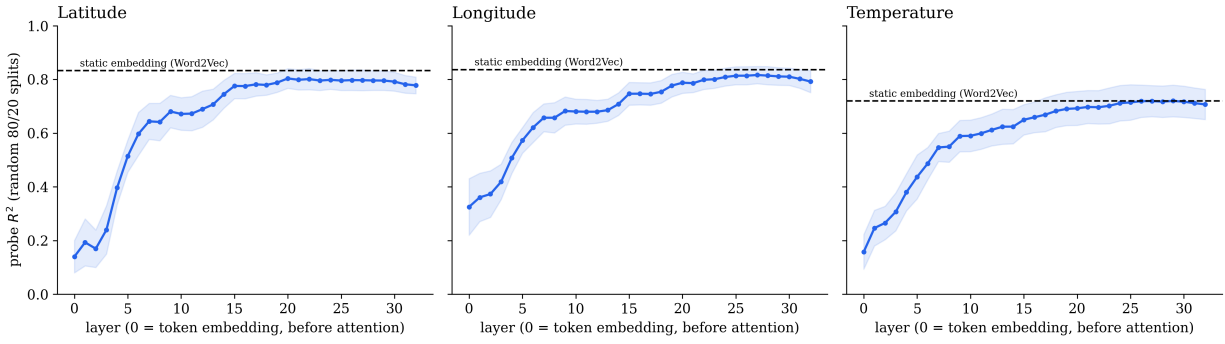


Figure 2: Probe R^2 by layer for Pythia-2.8B on the 272-city dataset under random 80/20 splits (shaded ± 1 s.d. over ten splits). Layer 0 is the token-embedding output, before any attention; 97% of these city names span several tokens, so layer 0 is the embedding of a final subword and the name has to be assembled by attention in later layers. The dashed line is the better static model (Word2Vec).

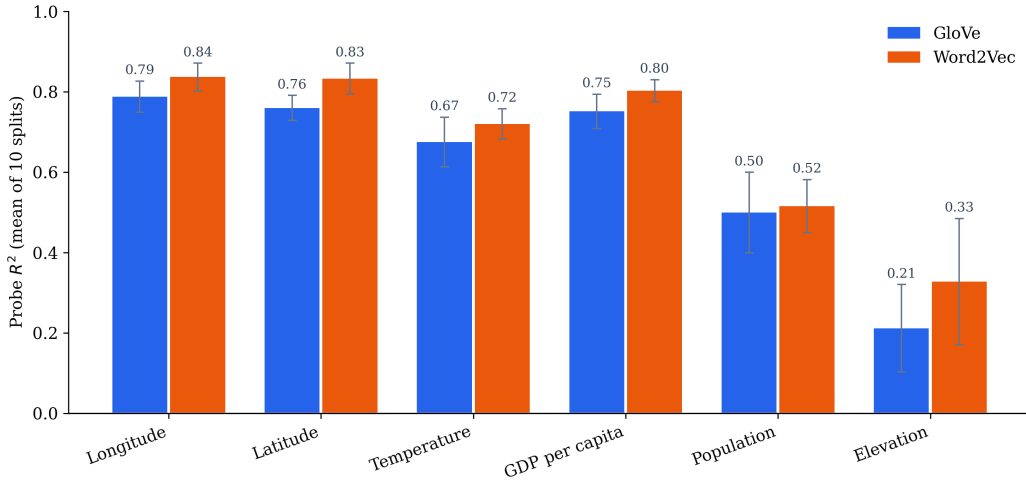


Figure 3: Probe R^2 for all six city targets, comparing GloVe and Word2Vec. Bars show the mean over ten random 80/20 train/test splits; error bars show ± 1 s.d. Geographic, climatic and national-economic targets yield high R^2 ; elevation is recovered weakly and with large variability across splits.

removes almost no variance and returns the full value exactly, confirming that the removal depends on the city correspondence rather than on the regression itself. What matters is not how much variance is removed but which.

The complementary condition is a consistency check rather than a finding: because the projection is a linear function of the static vectors, it contains no additional linearly available information. Separately regularized ridge fits can nevertheless differ slightly, because the transformation changes the effective penalty; empirically the two values are nearly identical (0.763 against GloVe’s own 0.760 for latitude; 0.837 against Word2Vec’s 0.833).

The two probes fail on the same cities. Signed prediction errors from the transformer probe and the static probe, correlated within each test fold with the target partialled out, give $r = 0.41$ – 0.58 across targets and static models, and $r = 0.27$ – 0.44 when computed within continent. The floor for this measure—the transformer probe against a probe on matched random features—is -0.02 to 0.08 . For calibration, the same measure applied to GloVe against Word2Vec, two representations that share a distributional basis, gives $r = 0.52$ – 0.62 .

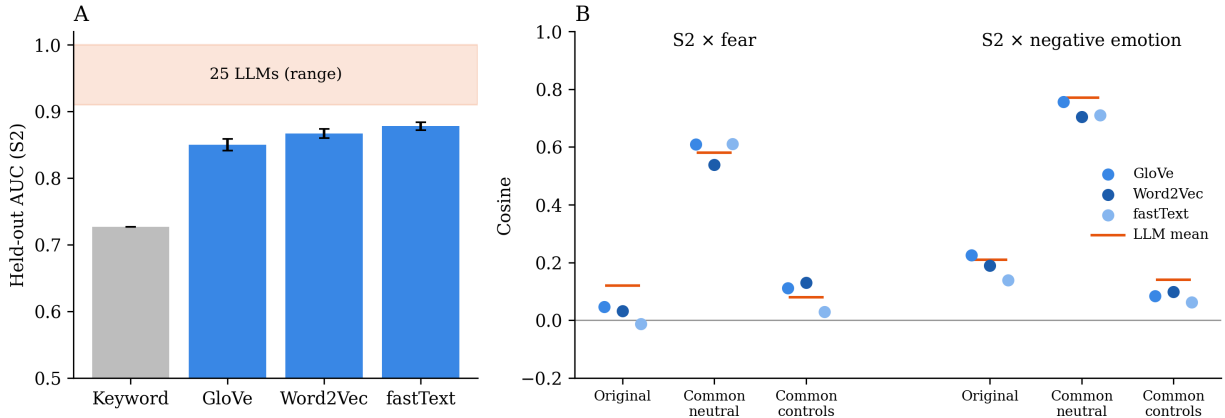


Figure 4: The representational tests reported for a “pain axis” in 25 LLMs (Tagliabue et al., 2026), run on averaged static word vectors of the authors’ own sentences. *A*: held-out separation of pain from controls (S2, first person; five-fold cross-validation over sentence sets, mean $\pm$ s.d. over 20 fold assignments). The keyword baseline is the best of several surface-lexical classifiers reported by an independent review (pain-axis-review contributors, 2026). LLM values are each model’s best layer. *B*: cosine of the pain direction with fear, negative-emotion and negative-world-state directions; bars span the LLMs’ per-model range.

Table 9: Removing the static-predictable component of Pythia-2.8B activations, in each split at the layer selected by cross-validation on that split’s training rows (Section 3.5); values are means over the ten test folds. *Var. removed* is the fraction of activation variance the removal takes out; the variance-matched null removes the same fraction along random, city-blind directions. The shuffled-static null repeats the removal with static vectors attached to the wrong cities.

Target	Static removed	full	projection	residual	var.-matched null	var. removed
Latitude	GloVe	0.800	0.763	0.233	0.791	34%
Longitude	GloVe	0.810	0.792	0.128	0.810	35%
Temperature	GloVe	0.711	0.684	0.220	0.711	35%
Latitude	Word2Vec	0.800	0.837	0.172	0.789	25%
Longitude	Word2Vec	0.810	0.833	0.129	0.809	28%
Temperature	Word2Vec	0.711	0.733	0.280	0.706	26%

Both show a modest relationship with neighbor availability. Probe error tends to rise as a city’s nearest training neighbors become less similar, in the static embeddings and, more weakly, in activation space. Each of the 239 cities that fall in at least one test set contributes one observation: its absolute error and the mean similarity of its five nearest training neighbors, each averaged over the splits in which it was tested, with continent partialled out. The correlation is $r = -0.16$ (latitude, $p = .01$), -0.15 ($p = .03$) and -0.05 ($p = .43$) in Pythia-2.8B activations, against -0.14 ($p = .03$), -0.27 ($p < .001$) and -0.19 ($p = .004$) in GloVe. The probes appear to answer partly by proximity to cities they were trained on, more clearly in GloVe.

A.9 What Carries the Recoverable Structure

Three analyses under the random 80/20 protocol ask how much of the recoverable structure is coarse group membership, and a fourth asks how well items are ordered within their group.

Six indicators outperform 300 dimensions. Replacing the embeddings with a six-element one-hot vector encoding continent membership and nothing else (Table 10) recovers longitude at $R^2 = 0.915$, above

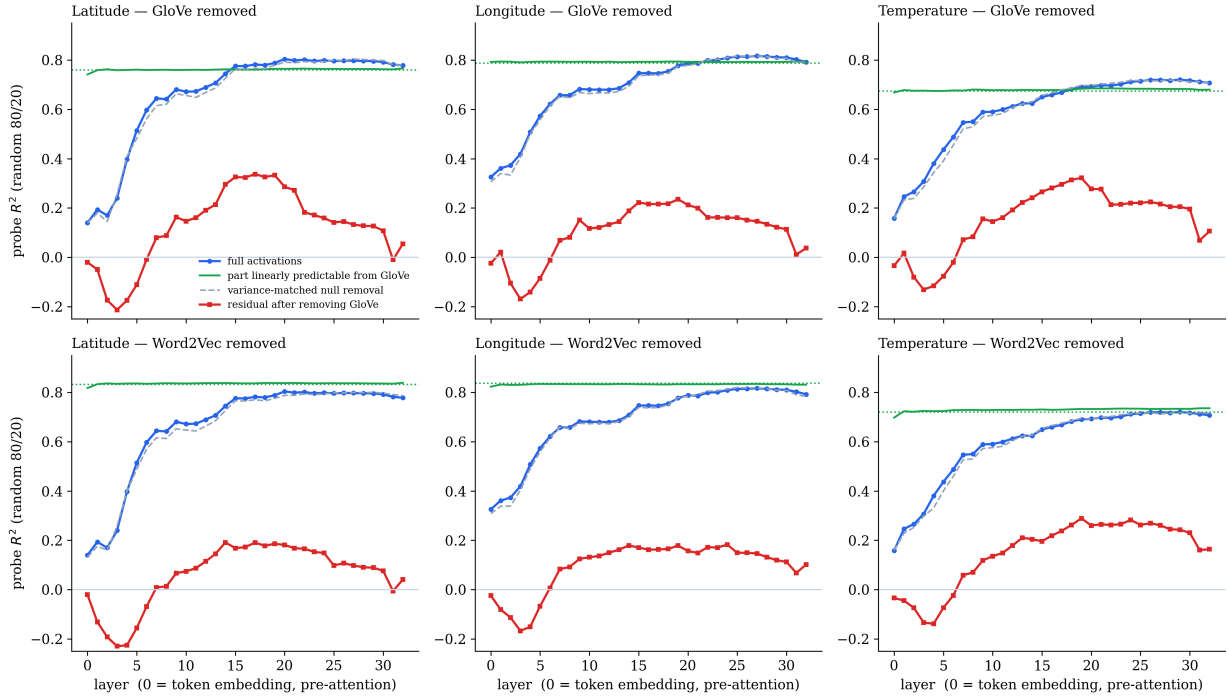


Figure 5: Pythia-2.8B by layer: full activations (blue), the component linearly predictable from the static embedding (green), the residual after that component is removed (red), and a removal of matched variance along random directions (gray dashed). The dotted line is the static embedding probed directly. Most probe-relevant performance lies in the activation component linearly predictable from static embeddings, while a nonzero residual remains; removing matched variance along random directions has little effect.

GloVe’s 0.788 and Word2Vec’s 0.837, and latitude at 0.731, close to GloVe’s 0.760. Temperature is the exception: continent identity alone yields only 0.290 against GloVe’s 0.675.

Recoverability tracks between-group variance. The share of each target’s variance that lies between continents (η^2) is 0.926 for longitude, 0.747 for latitude, 0.606 for GDP per capita, 0.352 for population, 0.327 for temperature and 0.109 for elevation. Across the six targets, random-split R^2 tracks η^2 at Pearson $r = 0.85$ and 0.87 for GloVe and Word2Vec (Spearman 0.94 for both); excluding temperature, $r = 0.94$ and 0.95 . Temperature is the residual, exceeding its η^2 -predicted value by $+0.232$ (GloVe) and $+0.207$ (Word2Vec).

This is a fact about the targets and the sample, not only about the representations: a variable whose variance is overwhelmingly between coarse groups can be recovered at high R^2 by anything that identifies the group, without any finer-grained information being present.

Decomposition. Substituting each city’s vector with its continent’s centroid (computed from training rows only) and, separately, with its within-continent residual isolates the two components under the random protocol. For GloVe, latitude decomposes as 0.760 (full), 0.731 (centroid only), $+0.011$ (residual); longitude as 0.788, 0.915, -0.018 ; temperature as 0.675, 0.287, $+0.242$. Word2Vec gives 0.833/0.731/ $+0.002$, 0.837/0.915/ -0.021 and 0.720/0.288/ $+0.315$. For the coordinates the centroid accounts for nearly all of the full R^2 ; for temperature it accounts for less than half.

Holding out continents. Training on five continents and predicting the sixth, with predictions pooled over the six held-out continents, GloVe and Word2Vec reach $R^2 = 0.003$ and 0.319 for latitude, -0.246 and -0.218 for longitude, and 0.476 and 0.465 for temperature, against 0.76 – 0.83 , 0.79 – 0.84 and 0.67 – 0.72 under random splits. Random-split coordinate performance depends heavily on continent-specific calibration, and

Table 10: Probe R^2 and within-continent ordering for the 272 cities under random 80/20 splits (mean over ten splits). Within-continent r correlates predictions with targets after removing continent means in each test set. The one-hot rows carry only country or continent identity (132 countries).

Representation	Target	R^2	Within-continent r
GloVe	Latitude	0.760	0.670
	Longitude	0.788	0.410
	Temperature	0.675	0.743
Word2Vec	Latitude	0.833	0.755
	Longitude	0.837	0.519
	Temperature	0.720	0.792
Country one-hot	Latitude	0.510	—
	Longitude	0.754	—
	Temperature	0.364	—
Continent one-hot	Latitude	0.731	0
	Longitude	0.915	0
	Temperature	0.290	0

the learned linear map transfers poorly to a continent it has not seen; temperature, much of whose variance lies within continents, transfers substantially better.

Within-group ordering. Correlating predictions with targets after removing continent means measures ordering inside continents alone, and a representation carrying only continent identity scores zero. Under random splits GloVe orders cities within continents at $r = 0.67$ for latitude, 0.41 for longitude and 0.74 for temperature, and Word2Vec at 0.76, 0.52 and 0.79. The embeddings therefore carry position within continents as well as continent identity, consistent with the dependence of similarity on distance (Appendix A.10). The decomposition above does not show this, because a probe trained on within-continent residuals is scored against the full target, most of whose variance lies between continents.

Historical figures. On the released figures, year of death varies almost entirely between centuries ($\eta^2 = 0.996$ in the test set) and mostly between three eras (before 500 CE, 500–1400, after 1400; $\eta^2 = 0.80$). As oracle decompositions derived from the target itself, an era label accounts for $R^2 = 0.80$ and a century label for 1.00, against 0.50 for GloVe. Replacing each figure’s GloVe vector with its era’s training centroid gives 0.80, and the within-era residual gives 0.04. Within centuries, GloVe and Word2Vec order figures at $r = 0.05$.

Other targets. Across all six city properties, random-split R^2 tracks η^2 . Elevation departs in the other direction: it is recovered worst under the random split, yet its recovery is least attributable to continent membership, with 63–82% of its modest recoverability surviving the removal of continent centroids, against 12–17% for GDP. Recoverability and group dependence are dissociable, so a high random-split R^2 does not by itself say what kind of information a representation carries.

A.10 Similarity Tracks Distance, Not Only Co-membership

A natural reading of the preceding results is that embedding similarity is driven by shared administrative and linguistic context rather than by geography as such. That is not the whole explanation. Across all 36,856 city pairs, cosine similarity falls with great-circle distance independently of co-membership. With log distance, same-country and same-continent indicators entered together, the standardized coefficient on distance is -0.375 (GloVe) and -0.409 (Word2Vec), against $+0.134$ and $+0.082$ for same-country membership. Distance continues to predict within every stratum: among same-country pairs $r = -0.480$ and -0.488 ; among same-continent, different-country pairs -0.388 and -0.353 ; among pairs on different continents -0.241 and -0.294 . Permuting city labels (10,000 draws) puts the partial distance coefficient far outside its null (-0.375 against

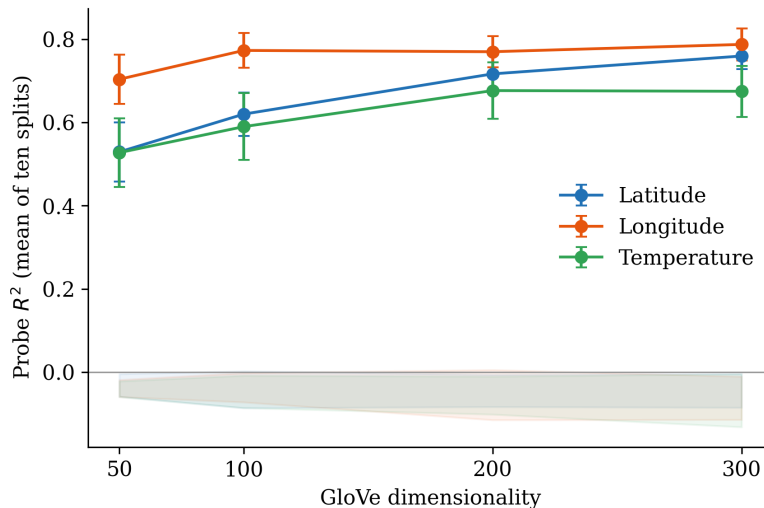


Figure 6: Probe R^2 versus GloVe embedding dimensionality for latitude, longitude, and temperature (272 cities). Points are means over the ten splits of Table 6, with error bars of ± 1 s.d.; shaded bands show ± 2 s.d. of matched random embedding baselines.

0.000 ± 0.020 , $p < .001$). Pairs under 500 km apart in *different* countries are more similar (GloVe 0.419, Word2Vec 0.503) than pairs over 2000 km apart within the *same* country (0.293, 0.364).

The representations therefore carry genuine local metric structure.

A.11 Dimensionality Scaling

To test whether the geographic signal scales smoothly with embedding capacity, we run ridge regression probes predicting latitude, longitude, and temperature from GloVe embeddings at four dimensionalities (50, 100, 200, 300) using the 272-city dataset, under the protocol of Table 6 (mean over the same ten random 80/20 splits).

Figure 6 shows that R^2 rises with dimensionality for all three targets, though longitude changes little beyond 100 dimensions and temperature beyond 200. Latitude R^2 rises from 0.53 (50d) to 0.76 (300d); longitude from 0.70 to 0.79; temperature from 0.53 to 0.67, with the 300d values matching Table 6. (An earlier version of this figure used a single split, which overstated temperature at 300d.) These results show an association between dimensionality and performance within the tested GloVe models. They do not establish how performance would scale beyond 300 dimensions, nor do they isolate dimensionality from the other differences between static embeddings and transformers—corpus size, contextual disambiguation, training objective and architecture all remain possible contributors. The dimension-matched comparison with Pythia in Appendix A addresses this more directly, by holding the number of features constant across representations.

A.12 Random Embedding Control

To confirm that the probes do not find spurious structure in arbitrary high-dimensional vectors, we generate 20 sets of random Gaussian embeddings at each dimensionality (50, 100, 200, 300) and run the same probes over the same ten splits. Figure 7 shows that random embeddings yield $R^2 \approx 0$ (slightly negative) at all dimensionalities: arbitrary vectors of the same dimensionality do not reproduce the observed performance, so the structure recovered from GloVe is not an artifact of high-dimensional geometry or of the fitting procedure. This control does not address effects of entity selection, regional clustering among the sampled cities, or preprocessing.

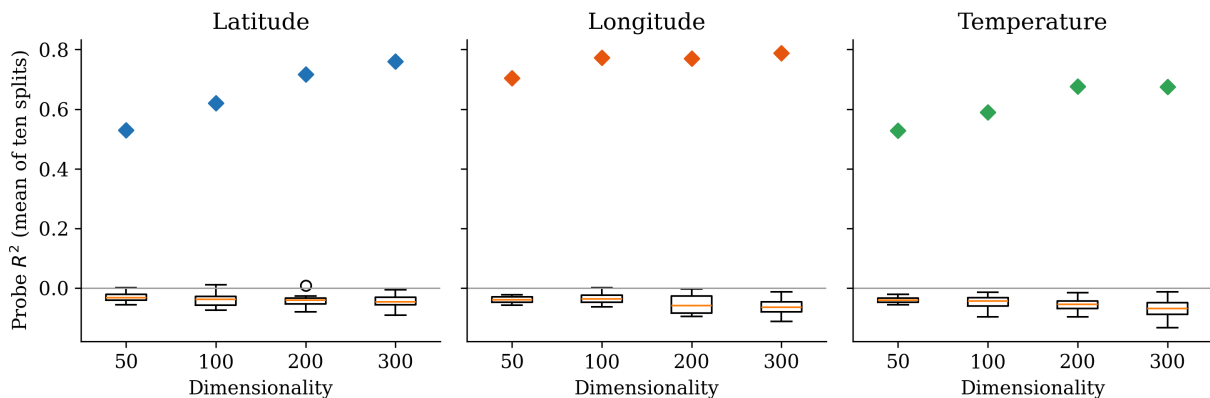


Figure 7: Random embedding control. Box plots show the ten-split mean probe R^2 for each of 20 sets of random Gaussian vectors at each dimensionality. Colored diamonds show corresponding GloVe R^2 values. The probes find no structure in random embeddings at any dimensionality.

A.13 Semantic Similarity Analysis

The preceding results establish that static embeddings preserve substantial spatial and temporal structure. We next ask whether this structure is semantically interpretable: does the embedding space encode geography through co-occurrence with meaningful vocabulary, and if so, which words carry the signal?

Data-driven word correlations. Rather than selecting probe words *a priori*, we take a data-driven approach. For every word in the GloVe vocabulary (restricted to the 20,000 most frequent words, excluding proper nouns, city and country names, and words shorter than four characters), we compute its cosine similarity to each of the 272 city embeddings, producing one similarity value per city. We then correlate these 272 similarity values with the cities’ actual mean annual temperatures (Pearson r). Figure 8 shows the 15 most positively and 15 most negatively correlated words.

Both directions are semantically interpretable. Words most associated with warmer cities reflect tropical ecology and disease burden—“dengue” ($r = +0.61$), “malaria” ($r = +0.59$), “jungle” ($r = +0.56$), “crocodile” ($r = +0.52$), “cyclone” ($r = +0.52$), “mosquito” ($r = +0.50$), “cholera” ($r = +0.48$). Words most associated with colder cities reflect winter activities and European cultural institutions—“skaters” ($r = -0.57$), “skating” ($r = -0.56$), “philharmonic” ($r = -0.53$), “violinist” ($r = -0.53$), “orchestra” ($r = -0.52$), “conservatory” ($r = -0.51$), “skiing” ($r = -0.50$), “conductor” ($r = -0.50$).

These semantic categories were not selected *a priori*; they emerge from an exhaustive search over the vocabulary. Because the reported words are selected for having large correlations, we treat the rankings as exploratory and report no inferential test on them. The results confirm that the embedding space encodes temperature through graded co-occurrence with climatically and culturally relevant vocabulary.

Shared company. To see why a city name’s representation is similar to those of words such as *malaria* or *orchestra*, we ranked each of the 50,000 most frequent GloVe words by its similarity to the mean representation of the names of the 25 warmest (or coldest) of our cities, taken relative to the mean over all cities, and separately by its similarity to the anchor words, and took the words that rank high on both. For warm cities and *malaria* and *jungle*, these are mostly Southeast Asian place names and tropical landscape (*jungles*, *borneo*, *cambodia*, *laos*, *mekong*, *thailand*). For cold cities and *skiing*, they are winter sports and Nordic places (*hockey*, *lillehammer*, *rink*, *finland*, *nordic*); for cold cities and *orchestra*, Russian classical music and its institutions (*bolshoi*, *prokofiev*, *kirov*, *leningrad*, *shostakovich*, *tchaikovsky*). Words for the weather rank far lower: *humid*, for example, ranks below 20,000th in similarity to the warm cities’ names.

For the released historical figures the same search is dominated by the makeup of the names rather than by period vocabulary: the words whose similarity to the representations of the figures’ names best tracks

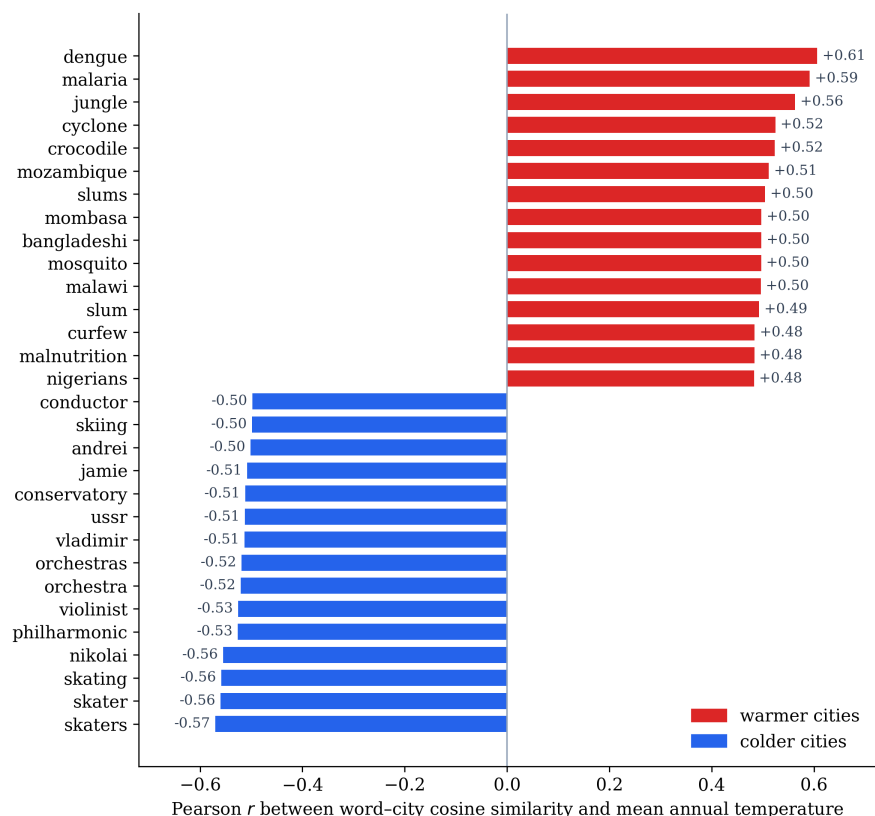


Figure 8: Data-driven identification of words whose GloVe embeddings correlate with city temperature. For each of the 17,250 common English words passing our filters, we compute cosine similarity to all 272 city embeddings and correlate with mean annual temperature. Shown are the 15 most positively correlated (red, associated with warmer cities) and 15 most negatively correlated (blue, associated with colder cities). Because the words are selected for extreme correlations from a search over the whole vocabulary, the figure is descriptive and no significance is implied. No words were selected *a priori*; the semantic categories emerge from the data.

year of death are Chinese provinces and cities (earlier; e.g. “henan”, “guangdong”, $r \approx -0.49$) and modern Western given names and surnames (later; e.g. “wally”, “carlson”, $r \approx +0.45$), along with occupations such as “chemist” and “biologist”. The temporal signal in the static vectors comes largely from where, and in what roles, people are written about.

Composite scores. The preceding analysis shows that geographic signal is distributed across many words in the vocabulary. A complementary question is how little information is needed: can just two well-chosen words capture the signal?

Composite scores constructed from antonym pairs suggest that they cannot. The cold–warm composite—defined as cosine similarity to “cold” minus similarity to “warm” for each city—correlates with latitude at only $r = 0.41$ and with temperature at $r = -0.34$. For the released historical figures, a modern–ancient composite correlates with year of death at only $r = 0.23$, consistent with the name-driven structure described above. Together with the data-driven analysis, these results indicate that the signal is distributed across much of the vocabulary rather than concentrated in the words for the property itself.

Table 11: Semantic subspace ablation (272 cities, GloVe). ΔR^2 : drop in ten-split mean R^2 when a category’s subspace is removed. z : the drop against 200 random word sets matched on number of words, frequency, proper-name status, vector norm and dimensionality; * marks an empirical one-sided $p \leq .01$ (1/201 is the smallest attainable). The last column is the share of the city embeddings’ variance the subspace removes, with the mean over the matched sets in brackets.

Category	Dims	Latitude		Longitude		Temperature		City variance removed
		ΔR^2	z	ΔR^2	z	ΔR^2	z	
Country names	20	+0.226	3.3*	+0.155	2.0	+0.224	3.2*	.29 (.24)
Region & continent	18	+0.119	2.3	+0.011	-1.6	+0.063	0.7	.16 (.17)
Climate & weather	19	+0.048	4.0*	+0.007	0.0	+0.090	7.7*	.12 (.11)
Cultural & language	20	+0.031	-0.4	+0.026	-0.7	+0.067	0.7	.20 (.18)
Cardinal directions	9	+0.018	0.6	+0.007	-0.9	+0.021	1.3	.07 (.08)
Economic terms	19	+0.004	-0.3	-0.000	-1.0	+0.015	0.8	.11 (.10)

A.14 Semantic Subspace Ablation

Table 11 reports the R^2 drop for each semantic category when its subspace is removed from the GloVe city embeddings, and its z -score against random word sets matched on number of words, frequency, proper-name status, vector norm and dimensionality (Appendix A.6). Baseline (unablated) R^2 values are 0.760 for latitude, 0.788 for longitude and 0.675 for temperature.

Against matched word sets, three results stand out. Removing **climate and weather** terms costs temperature far more than matched vocabularies do ($\Delta R^2 = 0.090$, $z = 7.7$), and latitude somewhat more ($z = 4.0$), but leaves longitude where matched sets do ($z = 0.0$), although these terms remove no more of the city embeddings’ variance than the matched sets. **Country names** produce the largest decrement on every target and exceed the matched sets for latitude and temperature ($z = 3.3$ and 3.2); their subspace also aligns more closely with the city vectors than any matched set (29% of the city variance against $24 \pm 1\%$). Regressing the matched sets’ decrements on the share of city variance they remove and extrapolating to the country subspace’s share predicts decrements of 0.15–0.16, which matches the longitude decrement and accounts for 35–43% of the excess on latitude and temperature; because the country subspace lies outside the range of the matched sets, this attribution is approximate. **Region and continent** names exceed the matched sets only for latitude ($z = 2.3$, $p = .03$). The remaining vocabularies (cultural and language terms, cardinal directions and economic terms) cost no more than matched random words. Against random orthonormal subspaces, the comparison used in an earlier version of this analysis, every category except economic terms appeared to have a large effect (z up to 42); most of that reflected the weakness of that null rather than anything specific to the vocabularies.

The data thus indicate that what the probes use is concentrated in directions shared with country names, and, for temperature, with words for climate and weather, in line with the word correlations above. Two limits should be stated. The ablations remove directions in embedding space; they are not interventions on the corresponding words in the training corpus, and each subspace may carry correlated information from several sources, so the individual drops are not additive contributions. And the category labels describe the vocabulary used to construct each subspace, not the content the subspace is thereby shown to contain.